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Deep learning surrogate model to explain decision variable synergies in energy system modelling

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ABSTRACT

Energy system models are powerful tools for planning the low-carbon transition. They identify optimal technology portfolios but rarely explain why certain combinations outperform others. Understanding decision variable synergies across multiple sectors remains a key gap in the literature. This study addresses that gap with a two-stage methodology applied to the Piemonte region (Italy) for 2050, using 17 decision variables and two objective functions (total annual costs and CO_2 emissions). In the first stage, an expert-based analysis of six electrification technologies is conducted using the EPLANopt model. It reveals a fundamental behavioral dichotomy: heat pumps reduce CO_2 independently of grid decarbonization, while Power-to-X technologies require an effective renewable share above approximately 50% to yield a net environmental benefit. In the second stage, a deep learning surrogate model trained on thousands EnergyPLAN simulations enables systematic sensitivity and interaction analysis across all 17 variables. Explainability artificial intelligence methods identify solar capacity and synthetic gas as the dominant variables. Pairwise analysis across all 136 variable combinations reveals a dominant PV surplus–Power-to-Gas valorisation chain. Third-order analysis shows that battery storage acts as a critical mediator between PV generation and Power-to-Gas conversion, producing emergent synergies that cannot be decomposed into pairwise effects. These findings provide a physically interpretable, data-driven basis for sequencing technology deployments in regional energy planning.

1. Introduction

Climate change is primarily driven by greenhouse gas emissions from energy sectors [1,2]. Energy system models have become essential tools for planning the transition to low-carbon energy systems [3,4]. Modern energy systems involve complex interactions across electricity, heating, transport, and industrial sectors [5]. Successfully decarbonizing these systems requires understanding how different technologies and interventions work together [6]. Traditional energy system optimization identifies which technology combinations minimize economic and/or environmental objective functions [7]. However, these approaches often reveal *what* the optimal solutions are but provide limited insight into *why* certain combinations outperform others [8]. Smart energy systems require coordinated planning across multiple sectors to integrate fluctuating renewable energy sources effectively [9]. Yet understanding the complex interactions between decarbonization measures across these integrated sectors remains a significant challenge [10]. This gap limits policymakers' ability to design robust transition strategies and sequence technology deployments effectively.

The aim is first to understand how the scientific literature has addressed the challenge of identifying and explaining correlations between the inputs, in particular decision variables, and outputs, objective function values, of energy system model results. We focus particularly on studies that moved beyond calculating optimal scenarios to systematically analyse why certain technology combinations perform better than others and how groups of technologies interact. Energy system models based on investment or capacity expansion optimization use decision variables, in order to minimize or maximize an objective (most often total system cost, or emissions), subject to constraints (energy balances, capacity limits, policy targets, etc.) [11]. Several studies have

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attempted to understand the relationships between these inputs (decision variables values) and outputs (objective functions values), though with varying degrees of methodological rigor.

Basic correlation-based sensitivity analysis provides the simplest approach to understanding decision variable influences on objective functions. Following capacity expansion optimization, several studies have analyzed the relationship between individual decision variables and objective functions using statistical correlation measures. This approach measures the strength and direction of the relationship between each decision variable and each objective function, allowing researchers to identify which variables are most strongly correlated with the output. For example, Conti et al. [12] (2019) employed Pearson's linear correlation coefficient method to quantify dependence between each design variable and each objective function and then reduce the optimization problem to variables with high correlation. In multi-objective optimization, other approaches examine how decision variables behave across the Pareto optimal solution set. Bellocchi et al. [13] used a rank-based visualization of variable co-variation along the Pareto front, which is closer in spirit to a monotonic (Spearman-type) ordering analysis and applied this approach to the Italian energy system, visualizing how decision variables, including renewable capacity, EV penetration, and fossil fuel consumption, vary across optimal trade-offs between costs and emissions. Importantly, none of these studies considered temporal dependence of decision variable influences, such as cross-correlations at time lags, as their analyses were performed on aggregated annual outputs rather than on time-series model inputs or intermediate variables. This reflects a broader limitation of static correlation approaches applied to the outputs of energy system simulations.

These approaches focus primarily on individual decision variables and do not systematically quantify whether specific combinations of technologies produce synergistic or antagonistic effects, that is, whether their combined impact differs from the sum of their individual contributions. However, because evaluating all possible combinations of decision variables is often computationally prohibitive, modellers typically restrict the analysis to selected groups of variables based on their domain knowledge and experience, following an expert-based approach. Connolly et al. [14] explored interactions between renewable energy technologies and storage options in 100% renewable energy scenarios through targeted scenario construction based on domain expertise. Similarly, Lund and Mathiesen [15] investigated the synergies between heat pumps and district heating systems by manually varying key parameters and interpreting results. Magni Johannsen et al. [16] demonstrated how different packages of decision variables, namely variable renewable energy sources in the electricity sector, heat pumps in district and individual heating sectors, and electrical storage technologies, can lead to progressive improvements in the two selected objective function indicators: total annual costs and annual CO_2 emissions. These approaches provide valuable insights but rely heavily on researcher intuition to select which variable combinations to explore. While systematic exploration of different decision variable combinations and their impacts on objective functions have traditionally been computationally prohibitive, recent advances in artificial intelligence techniques, particularly surrogate modelling, now enable emulation of complex energy system models with dramatically reduced computational time, opening pathways for comprehensive analysis of technology interactions across the full decision space [17]. Deep learning surrogate models [18], also termed metamodels or emulators [19], are simplified mathematical approximations of complex computational models. Usually based on Artificial Intelligence (AI) methods, they are trained to reproduce the input-output behavior of the original complex model using a limited set of simulation runs. Once trained, surrogate models can generate predictions almost instantaneously compared to the hours or days required by the original simulations. This makes them valuable tools for exploring large parameter spaces and conducting extensive uncertainty analyses that would otherwise be computationally prohibitive [20].

Table 1 summarizes the methodological landscape of decision variable interaction analysis in energy system modeling studies. The table characterizes each contribution along two axes: model features and analytical depth. Model features include the underlying energy system framework, its temporal and spatial resolution, and the sectors covered. Analytical depth is captured through five progressive dimensions: (i) whether the study applied an expert-based sensitivity analysis on pre-selected groups of decision variables, specifying which technology clusters were investigated; (ii) whether a systematic global sensitivity analysis was performed, quantifying how each individual decision variable independently affects the objective functions (first-order interaction); (iii) whether interactions between groups of two or more decision variables were explored, revealing how their joint deployment contributes to objective functions beyond the sum of individual effects (second- and higher-order interactions); (iv) whether a systematic ranking of synergistic and antagonistic technology combinations was carried out; and (v) whether Explainable AI (XAI) methods were applied to interpret the relationships between decision variables and objective functions.

Studies relying exclusively on expert-based approaches, Bellocchi et al. [13], Johannsen et al. [16], and Prina et al. [21], provide physically grounded insights into selected technology-RES interactions but are inherently limited to the variable combinations chosen by domain experts, leaving higher-order and non-intuitive interactions undetected. Studies employing machine learning surrogates, Neumann and Brown [22], Jahangiri et al. [24], and Quick et al. [23], achieve global coverage of the decision space, with Jahangiri et al. [20] additionally applying XAI techniques to interpret surrogate predictions. However, none of these studies advances beyond second-order interaction analysis, and none combines systematic synergy quantification with XAI-based interpretability within a multi-sectoral energy system framework. The present study addresses all three gaps simultaneously: it integrates expert-based and surrogate-enhanced exploration, extends interaction analysis to third order, introduces an explicit synergy ranking procedure, and applies both SHAP and Sobol-based XAI methods to provide physically interpretable explanations of decision variables interdependencies across all sectors.

The study is scoped to the Piemonte region in Italy, chosen for its representativeness of a highly industrialized, European regional energy system under active policy-driven transition. The analysis considers the target year of 2050, consistent with the Regional Energy and Environmental Plan (Piano Energetico Ambientale Regionale - PEAR) and the European Green Deal decarbonization targets. The methodology is designed to be readily applicable to other regional and national energy system contexts.

The remainder of the paper is organized as follows. Section 2 describes the materials and methods, including the EPLANopt modeling framework, the expert-based exploratory methodology, the surrogate model development and validation procedure, and the XAI-based interaction analysis. Section 3 presents the Piemonte regional case study, including the 2021 baseline characterization and the full set of decision variables and exogenous parameters. Section 4 reports the results of both the expert-based exploration and the surrogate-enhanced analysis, including model performance metrics, Pareto front validation, SHAP and Sobol analyses, and multi-order interaction results. Section 5 presents the conclusions and directions for future research.

2. Materials and methods

EnergyPLAN, developed by Aalborg University since 1999, implements smart energy systems by integrating multiple energy sectors and identifying synergies between them [25]. Lund et al. [26] introduced the Smart Energy Systems concept [5], formally defined as "an approach in which smart electricity, thermal and gas grids are combined with storage technologies and coordinated to identify synergies between them in order to achieve an optimal solution for each individual sector as well as

Table 1
Comparative overview of literature on decision variable interaction methodologies in energy system modeling studies. DV: Decision Variables.

Authors	Energy system model	Time resolution	Spatial resolution	Sectors included	Expert based DVs sensitivity analysis	Global DVs sensitivity analysis	Max interaction order	Systematic synergy ranking	XAI interpretability
Bellocchi et al. [13] 2020	EPLANopt (Italy)	Hourly	Single-node	Power, Heat	✓ (DVs about Electrification)	-	-	-	-
Johannsen et al. [16] 2023	EPLANopt (municipal level)	Hourly	Single-node	Power, Heat	✓ (RES, storage systems, heating technologies)	-	-	-	-
Prina et al. [21] 2024	EPLANopt (Italy)	Hourly	Single-node	All sectors	✓ (DVs about Electrification)	-	-	-	-
Neumann and Brown [22] 2023	PyPSA (Europe)	2h	128 nodes	power	-	✓ (Multi-fidelity approach)	1st	-	-
Jahangiri et al. [18] 2023	Copper (Canada)	Rapr. days	13 nodes	power	-	✓ (Deep Learning surrogate model)	2nd	-	-
Quick et al. [23] 2024	Balmorel (North of Europe)	Hourly	28 nodes	All sectors	-	✓ (Deep Learning surrogate model)	1st	-	-
Jahangiri et al. [20] 2024	Copper (Canada)	Rapr. days	13 nodes	power	-	✓ (Deep Learning surrogate model)	2nd	-	✓
Jahangiri et al. [24] 2024	Copper (Canada)	Rapr. days	13 nodes	power	-	✓ (Deep Learning surrogate model)	2nd	-	-
This article	EPLANopt (Piemonte region)	Hourly	Single-node	All sectors	✓ (DVs about Electrification of sectors)	✓ (Deep Learning surrogate model)	3rd	✓	✓

for the overall energy system" [27]. The model enables holistic cross-sectoral analysis encompassing buildings, industry [28], and transport [9], and supply technologies through electricity, gas, district heating [29] and cooling grids [6]. EnergyPLAN [30] performs hourly techno-economical simulations for systems based on renewable, fossil, and nuclear energy sources. Østergaard et al. [31] validated its effectiveness through a 2022 review documenting 315 peer-reviewed applications. The model analyzes key performance indicators including Primary Energy Supply, CO_2 emissions, excess power, and costs [32].

Numerous applications demonstrate EnergyPLAN versatility: single-objective optimization [33], Pareto front generation [34], and sector-specific applications in transport electrification, heating [13], and industrial decarbonization [35]. Recent developments include integration with multi-criteria decision analysis [36], island systems [37] and maritime decarbonization [38], economic impact assessment [39], and meta-heuristic optimization [40]. EPLANopt couples EnergyPLAN with Multi-Objective Evolutionary Algorithms (MOEAs) for expansion capacity optimization [7]. Prina et al. [41] extended EPLANopt for diverse applications including flexibility constraints [42], national [43] and regional scenarios [44], sector coupling [7], marginal abatement cost curves [45], supply risk assessment [46], and near-optimal solution evaluation [47].

This study employs EnergyPLAN version 16.1 with technical simulation strategy and dump charge for electric mobility. The MOEA [48] minimizes two objectives: total annual costs and annual CO_2 emissions. Decision variables (DV) are constrained by predetermined bounds. Equation (1) presents the objective functions of the multi-objective minimization problem. The optimization is subject to a set of constraints, the primary ones enforcing that each decision variable remains within a prescribed range defined by its lower $DV_k^{(L)}$ and upper $DV_k^{(U)}$ bounds. Additional constraints, such as the balance between energy demand and generation at each time step, as well as storage dynamics with identical initial and final states of charge, are implemented within the EnergyPLAN software.

$$\begin{aligned} &\text{Optimization function} \quad \min_{DV} \begin{bmatrix} \textit{Total Annual Costs}\ [M€] \\ \textit{Annual } CO_2 \textit{ Emissions}\ [Mt] \end{bmatrix} \\ &\text{Subject to} \qquad DV_k^{(L)} \le DV_k \le DV_k^{(U)} \end{aligned} \tag{1}$$

The EnergyPLAN software is responsible for performing the annual operational simulations, while the MOEA is used to optimize expansion capacities [7]. The optimization process employs standard genetic algorithm operators, including selection, crossover, and mutation, and continues iteratively until convergence is achieved and the final Pareto front is obtained.

This paper adopts a two-stage methodology to progressively expand the scope of decision variable interaction analysis. The first stage (Section 2.1) implements a traditional expert-based exploratory approach on strategically selected combinations of technologies. The second stage (Section 2.2) introduces a deep learning surrogate modeling framework that dramatically reduces computational time per simulation, enabling systematic exploration of a significantly larger number of decision variable combinations. This surrogate-enhanced methodology overcomes the computational bottleneck inherent in expert-based approaches, facilitating comprehensive mapping of the multi-dimensional decision space and revealing technology interactions that would remain hidden under traditional computational constraints.

2.1. Expert-based exploration of decision variable synergies

This section adopts an expert-based exploratory approach with a specific objective: to evaluate the impact on the two objective functions (total annual costs and annual CO_2 emissions) of specific combinations of decision variables identified through domain expertise by the authors. Rather than exploring all possible variable combinations, which would be computationally prohibitive with the original complex EnergyPLAN energy system model, this methodology strategically focuses on understanding how key electrification technologies interact with renewable energy penetration in the electricity sector. The expert-based selection of decision variable combinations follows a clear rationale rooted in sector coupling logic. Each analyzed combination pairs: i) primary electrification technologies representing critical decarbonization levers in heating, industry, or power-to-X sectors, ii) variable renewable energy sources (RES) in the electricity sector, iii) supporting infrastructure held at maximum capacity to evaluate the primary technology under optimal enabling conditions.

All simulations in this expert-based exploration phase employ the original EnergyPLAN energy system model described in Section 2, ensuring full physical accuracy and hourly temporal resolution. This distinguishes the current section from Section 2.2, where a surrogate model replaces EnergyPLAN to enable higher computational efficiency and a broader exploration of decision variables combinations.

To rigorously quantify renewable energy integration and its interaction with electrification technologies, two complementary metrics were defined. These indicators provide distinct perspectives on the role of renewable energy sources within the system. The first metric, RES Share of Electric Demand, RES_share^{el}, quantifies the total renewable electricity generation relative to system demand, Eq. (2).

$$RES\_share^{el} = \frac{E^{RES}}{D^{el}} \cdot 100 \tag{2}$$

where E^{RES} represents the total annual electricity generated from renewable sources (TWh) and D^{el} denotes the total annual electricity demand (TWh). This metric can exceed 100% to account for scenarios with structural overcapacity, thereby enabling the evaluation of system flexibility and Power-to-X integration potential across the full spectrum of renewable deployment, from baseline conditions to extreme overgeneration scenarios. The second metric, Effective Share of Electric Demand, $RES\_share^{el,effective}$, measures the fraction of demand actually covered by renewable sources, excluding any surplus generation, Eq. (3).

$$RES\_share^{el,effective} = \frac{\sum_i \min\left(E_i^{RES}, D_i^{el}\right)}{D^{el}} \cdot 100 \tag{3}$$

where E_i^{RES} represents the electricity generated from renewable sources in timestep i and D_i^{el} denotes the electricity demand in timestep i.

To evaluate the effectiveness of different decarbonization technologies and their interactions with renewable energy penetration, two key performance indicators are employed: $Relative\_CO_2\_change$ [%] and $Relative\_Costs\_change$ [%]. These indicators quantify the marginal contribution of each technology by comparing system performance with and without the technology under investigation, while maintaining constant renewable energy supply levels.

$$Relative\_CO_2\_change = \frac{CO_2^{with} - CO_2^{without}}{CO_2^{baseline}} \tag{4}$$

where:

- CO_2^{with}: represents emissions when the selected decision variable is introduced into the system.
- $CO_2^{without}$: represents the emissions of the system calculated when the selected variable is not considered, but keeping unaffected the renewable penetration.
- $CO_2^{baseline}$: represents the emissions of the starting scenario (baseline).

$$Relative\_Costs\_change = \frac{Costs^{with} - Costs^{without}}{Costs^{baseline}} \tag{5}$$

where:

- $Costs^{with}$: represents emissions when the selected decision variable is introduced into the system.
- $Costs^{without}$: represents the emissions of the system calculated when the selected variable is not considered, but keeping unaffected the renewable penetration.
- $Costs^{baseline}$: represents the emissions of the starting scenario (baseline).

2.2. Surrogate-enhanced exploration of decision variable synergies

The expert-based exploratory approach detailed in Section 2.1 provides insights into specific technology-RES interactions but faces an inherent computational constraint: systematic exploration of multiple decision variables simultaneously requires thousands of EnergyPLAN simulations, each consuming several seconds. This computational bottleneck limits the feasibility of comprehensive multi-dimensional analysis across all 17 decision variables and prevents systematic quantification of synergistic or antagonistic interactions between multiple technologies. To overcome this limitation, this section introduces a deep learning surrogate modeling framework that emulates EnergyPLAN model reducing the computational time from seconds to milliseconds per evaluation. This computational speedup enables two advances: (1) training on datasets spanning the full decision space to learn system behavior comprehensively, and (2) performing systematic sensitivity analysis across all decision variable combinations to identify technology interactions that would remain computationally inaccessible using the original EnergyPLAN model. Section 2.2.1 details the surrogate model development, including training data generation, machine learning architecture selection, and performance validation. Section 2.2.2 presents the methodology for comprehensive decision variable interaction analysis enabled by the surrogate, incorporating explainable AI techniques to interpret model predictions and extract physically meaningful insights about technology synergies.

2.2.1. Surrogate-enhanced exploration of decision variable synergies

The surrogate model learns to emulate EnergyPLAN's input-output mapping by training on a comprehensive dataset of simulated energy system configurations. The training dataset is generated through systematic sampling of the decision space defined by the 17 decision variables listed in Table 2. Input features consist of the decision variable values for each simulated scenario. Each decision variable is constrained within its lower bound $\boldsymbol{dv}_{\boldsymbol{k}}^{(\boldsymbol{L})}$ (baseline 2021 value) and upper bound $\boldsymbol{dv}_{\boldsymbol{k}}^{(\boldsymbol{U})}$ as specified in Table 2. Output targets are the two objective functions computed by EnergyPLAN for each scenario: Total annual costs [M€], Annual CO_2 emissions [Mt]. To ensure uniform coverage of the high-dimensional decision space, training samples are generated using Sobol sequences, a low-discrepancy quasi-random sampling technique that distributes points more evenly than pure Monte Carlo sampling. This approach minimizes clustering and gaps in the sampled space, ensuring the surrogate learns system behavior across all regions of the decision domain.

For each sampled combination of decision variables, EnergyPLAN performs a complete annual simulation with 8784 hourly timesteps, calculating energy balances, storage dynamics, and economic-environmental performance. The computational cost of this data generation phase is substantial but performed once; the resulting dataset then enables rapid surrogate-based exploration. Prior to training, all input features and output targets undergo standardization (zero mean, unit variance) to prevent variables with larger numerical ranges from dominating the learning process. The dataset is partitioned into training (80%) and test (20%) sets, with the test set withheld entirely during model development to enable unbiased performance assessment. Multiple machine learning architectures were systematically evaluated to identify the model best suited for surrogate modeling of this multi-sectoral energy system. The supplementary material provides the full benchmarking results across all 16 evaluated architectures, their learning curves, and performance tables at each training fraction. The

Table 2

List of decision variables per sector and type, their current state $dv_k^{(L)}$ and their maximum potential $dv_k^{(U)}$.

Sector	Type	Name	Unit	Current state, $dv_k^{(L)}$	Maximum potential, $dv_k^{(U)}$
Power sector	Generation source	Residential Rooftop PV	MW el	233.2	12722.8
		Non-residential Rooftop PV	MW el	1020.8	6739.5
		Ground-mounted PV with Single-axis Tracker	MW el	537.5	866.9
		Agrivoltaics	MW el	0	73013.0
		Onshore Wind Power	MW el	18.8	1000
	Balancing & storage	Electric Storage (Lithium-ion Batteries)	GWh el	0	100
		E-methane for Gas Grid Injection	TWh th	0	20
		Biomethane for Gas Grid Injection	TWh th	0	10
		Electrolyzer Capacity for Hydrogen generation	MW th	0	20000
		Hydrogen Storage	GWh th	0	200
Heating sector	Generation & storage sources	Heat Pumps for District Heating	MW th	0	5000
		Thermal Storage for District Heating	GWh th	0	10
		Heat Pumps for Individual Heating	%	0	100
	Energy refurbishment	Energy efficiency of individual buildings	%	0	70
Industry sector	Generation source	Industrial Heat Pumps for Low-Temperature Processes	%	0	100
		Electric Furnaces for High-Temperature Processes	%	0	100
		Hydrogen for High-Temperature Industrial Processes	%	0	100

architecture was trained on the full 40,000-scenario dataset using an 80/20 train–test split, with all inputs and outputs independently standardized to zero mean and unit variance prior to training. Each model was trained on progressively larger subsets of the training data to construct learning curves, plots of prediction error versus training set size. This analysis serves two purposes: (1) identifying the minimum dataset size beyond which performance saturates (diminishing returns to additional data), and (2) comparing model convergence behavior to select the architecture with optimal bias-variance tradeoff for this application.

Surrogate model performance is quantified using complementary metrics that assess different aspects of prediction quality. Pointwise prediction accuracy is measured through the Coefficient of determination (R^2), Eq. (6).

$$R^2 = 1 - \frac{\sum_{m=1}^{M}(y_m - \widehat{y}_m)^2}{\sum_{m=1}^{M}(y_m - \overline{y}_m)^2} \tag{6}$$

where y_m represents the actual value calculated by EnergyPLAN for scenario m, $\widehat{y}_m$ is the surrogate model prediction, and $\overline{y}_m$ is the mean of actual values. Values approaching 1 indicate near-perfect reproduction of EnergyPLAN behavior. Pointwise accuracy is further quantified by the Root Mean Square Error (RMSE), which complements R^2 by penalizing large individual prediction errors more heavily than the mean absolute error and is particularly sensitive to outliers near the Pareto front (see the supplementary material).

Pareto front accuracy, critical for multi-objective optimization applications, is assessed using Generational Distance (GD), Eq. (7), and Inverted Generational Distance (IGD), Eq. (8), measuring the average distance from surrogate-predicted Pareto solutions to the true Pareto front obtained via EPLANopt.

$$GD(A) = \frac{1}{|A|}\cdot\left(\sum_{m=1}^{|A|} d_m^p\right)^{\frac{1}{p}} \tag{7}$$

where A is the set of solutions predicted by the surrogate, d_m^p is the Euclidean distance from scenario solution m in A to the nearest point on the reference Pareto front Z, and $p = 2$.

$$IGD(A) = \frac{1}{|Z|}\cdot\left(\sum_{m=1}^{|Z|} \widehat{d}_m^p\right)^{\frac{1}{p}} \tag{8}$$

where $|Z|$ is the number of points in the reference Pareto front and $\widehat{d}_m^p$ is the distance from each reference point to the nearest surrogate prediction. To verify that Pareto front reproduction quality is not an artefact of a single optimization run, the EPLANopt algorithm was coupled to the surrogate and executed across multiple independent runs with different random seeds. Full learning curves, the complete benchmarking table across all 16 architectures at three training fractions, and the multi-run convergence statistics are reported in the Supplementary Material.

2.2.2. Comprehensive variable interaction analysis

Once validated, the surrogate model enables systematic exploration of decision variable interactions at computational scales infeasible with the original EnergyPLAN model. This section details the methodology for quantifying how combinations of decision variables jointly influence the objective functions. Unlike the expert-based approach (Section 2.1) which examines pre-selected two-dimensional technology-RES pairings, the surrogate framework enables i) full decision space sampling, generation of hundreds of thousands of scenarios spanning all 17 decision variables simultaneously, exploring regions of the decision space, ii) high-order interaction detection through the identification of synergies and antagonisms involving three or more technologies acting in combination, iii) conditional sensitivity analysis through the quantification of how one technology's effectiveness varies as a function of other technologies' deployment levels.

The systematic analysis proceeds by generating dense grids across selected decision variable subspaces while marginalizing over others, effectively producing high-resolution multi-dimensional sensitivity surfaces. For example, to examine the three-way interaction between photovoltaic, electric storage, and heat pumps for district heating, a three-dimensional grid is constructed with each dimension discretized at fine intervals, the surrogate evaluates all grid points in seconds, and the resulting response surfaces reveal how storage and heat pumps modulate photovoltaic integration effectiveness.

To quantify how combinations of decision variables jointly influence system performance beyond what individual deployments would suggest, a synergy indicator is introduced. For a cluster of n decision variables, the CO_2 synergy effect is defined in Eq. (9).

$$\Delta CO_2^{\text{synergy}}\ [MT] = \left(CO_2^{group} - CO_2^{baseline}\right) - \sum_{j=1}^{J}\left(CO_{2,j} - CO_2^{baseline}\right) \tag{9}$$

where CO_2^{group} denotes the annual emissions obtained when all j decision variables of the cluster are simultaneously deployed, $CO_{2,j}$ is the emission level when only variable j is active and all remaining variables are held at their lower bounds, and $CO_2^{baseline}$ represents the reference scenario with all decision variables at their lower bounds. Each parenthetical term thus isolates the marginal emission reduction attributable to the group or to each individual variable relative to the same baseline. Synergy magnitudes are relative to this baseline, $CO_2^{baseline}$, and would differ quantitatively under alternative references (e.g., a mid-range scenario). The indicator is therefore a measure of non-linear interaction structure rather than an absolute policy-independent quantity.

A negative $\Delta CO_2^{synergy}$ indicates that the joint deployment achieves a greater total reduction than the sum of the individual contributions, a synergistic interaction, whereas a positive value signals an antagonistic relationship in which technologies partially undermine each other's effectiveness. An analogous indicator, $\Delta Costs^{synergy}$, is defined identically by replacing CO_2 emissions with total annual system costs, enabling the same interaction decomposition on the economic objective function. However, for simplicity the synergy analysis is performed only on annual CO_2 emissions. As the different SHAP importance rankings for costs versus emissions in Fig. 6 suggest, cost synergies may follow different interaction structures, and the emission-based results should not be generalized to the economic objective; extending this analysis to total annual costs is left for future work.

With the increase of the number of variables simultaneously studied, like pairs, triples, or higher-order combinations, the number of simulations required exponentially grows and can become incompatible with research constraints even for surrogate models. To manage this complexity, an Interaction Impact Score (I_{score}) is developed to evaluate and rank technological clusters, allowing for the a priori rejection of combinations with minor non-linear dynamics. The score is computed as a weighted combination of three distributional statistics of $\Delta CO_2^{synergy}$ evaluated over the full discretized subspace of the cluster, Eq. (10). The three statistics capture complementary aspects of the interaction structure. The peak magnitude, $\max\left(\left|\Delta CO_2^{\text{synergy}}\right|\right)$, identifies the strongest interaction effect anywhere in the subspace, whether synergistic (negative) or antagonistic (positive); the absolute value is necessary because synergistic clusters produce negative $\Delta CO_2^{\text{synergy}}$ values and would otherwise be ranked below zero-interaction clusters by a sign-sensitive maximum. The effect spread, $\max\left(\Delta CO_2^{\text{synergy}}\right) - \min\left(\Delta CO_2^{\text{synergy}}\right)$, measures how strongly the interaction varies as technology deployment levels change, capturing clusters whose synergy switches sign or amplitude across the decision subspace. The variability, $\sigma\left(\Delta CO_2^{\text{synergy}}\right)$, is the standard deviation of the interaction across all grid points, differentiating clusters with a spatially diffuse, moderate interaction from those exhibiting abrupt inflection points concentrated in a small region.

$$\begin{aligned} I_{\text{score}} = 0.6 \cdot \max(|\Delta CO_2^{\text{synergy}}|) + 0.2 \cdot (\max(\Delta CO_2^{\text{synergy}}) \\ - \min(\Delta CO_2^{\text{synergy}})) + 0.2 \cdot \sigma(\Delta CO_2^{\text{synergy}}) \end{aligned} \tag{10}$$

The three terms capture complementary aspects of the interaction: the peak absolute deviation identifies the maximum synergistic or antagonistic effect anywhere in the decision space; the range quantifies how strongly the interaction varies with technology deployment levels, reflecting its sensitivity to operating conditions; and the standard deviation differentiates clusters with a diffuse, moderate interaction from those exhibiting abrupt inflection points. Clusters are ranked in descending order of I_{score}, and only the highest-ranking subsets are subjected to detailed response-surface analysis. The procedure is applied recursively: the most influential lower-order clusters serve as the foundation for constructing and screening higher-order interactions. Alternative weighting configurations were also tested, including equal weights and schemes placing greater emphasis on the effect range or standard deviation term; in all cases, the top-ranked pairs and triplets remained substantially unchanged.

The surrogate model, once validated, constitutes an accurate and computationally efficient emulator of EnergyPLAN, but its predictive function remains opaque: it maps inputs to outputs through a high-dimensional non-linear transformation that does not, by itself, reveal why certain technology combinations outperform others. Explainable Artificial Intelligence (XAI) [49] refers to a body of methods designed to make the internal behaviour of such black-box models interpretable to human analysts [50]. Two complementary XAI frameworks are employed here, each probing a distinct dimension of the surrogate's behaviour. I) SHapley Additive exPlanations (SHAP) [51] decompose each individual surrogate prediction into additive attribution values, one per input variable, grounded in cooperative game theory [52]. Applied across the full set of sampled scenarios, SHAP values quantify both the direction and magnitude of each decision variable's marginal contribution to costs and emissions relative to the expected model output. Summary plots of the SHAP value distributions are used to rank variables by global importance and to identify monotonic or non-monotonic response patterns. II) Sobol Sensitivity Analysis. Variance-based global sensitivity analysis, as formalised by Sobol [53] and extended by Saltelli et al. [54], decomposes the total output variance into contributions attributable to individual variables and to their interactions. The first-order Sobol index measures the fraction of variance explained by each variable in isolation, while total-order indices additionally capture all higher-order interaction effects involving that variable. The gap between total-order and first-order indices therefore quantifies the degree to which a variable's influence is contingent on the simultaneous deployment of other technologies, a variance-based counterpart to the synergy indicator $\Delta CO_2^{\text{synergy}}$ defined above. Sobol indices are estimated using the Saltelli quasi-random sampling scheme [54] applied to the surrogate, making the variance decomposition computationally tractable across all 17 decision variables.

3. Case study: Piemonte region

This study builds upon the comprehensive energy system model for Piemonte region developed in our previous work [55]. The baseline energy system (year 2021) was constructed using EnergyPLAN software [30], integrating data from multiple sources. Regional energy statistics came from the Regional Energy and Environmental Plan (PEAR) [56] and annual Statistical Energy Reports [57] produced by the Sustainable Energy Development Office. National data sources included GSE [58], ENEA [59], MASE [60] and Terna [61]. Electricity generation profiles were based on GSE renewable energy statistics, while thermal demand patterns were derived from regional building energy performance and thermal systems registries. Industrial energy consumption was characterized by temperature requirements and process types through analysis by Politecnico di Torino's Energy Center. District heating data came from network operators, particularly Iren [62] for the Turin metropolitan area. Solar generation profiles were developed using PVGIS data [63] with regional adjustments. Technology costs, performance parameters, and emissions factors were aligned with 2021 market conditions, with projections for 2030, 2040, and 2050 based on established learning curves and technology development trajectories [55]. Detailed graphical representations of the Piemonte energy balance by fuel type, including the evolution of the energy mix and projected growth rates across the 2030, 2040, and 2050 horizons, are reported in Ref. [55], where the regional case study is presented and discussed in full.

Decision variables represent decarbonization measures whose sizes are optimized in the model. The final set of decision variables was developed iteratively through collaboration with the regional authority to address specific priorities and constraints [55]. Each variable has a lower bound (current state within the baseline 2021) and an upper bound (maximum technical potential), see Table 2. The upper bounds were derived through different approaches depending on the technology type: for PV technologies, including agrivoltaics, the maximum potential was estimated through a GIS-based spatial analysis that identified suitable areas after applying exclusion layers for hydrogeological risk, landscape protection, high-quality agricultural land, and other territorial constraints, then calculated installable capacity using technology-specific ground coverage ratios and system efficiencies; the agrivoltaic upper bound of 73,013 MW corresponds to the more permissive land-use scenario allowing installations across all suitable agricultural land not subject to constraints, representing a physical ceiling on installable capacity rather than a deployment target [55]. For electric storage, the 100 GWh cap was determined through preliminary simulations showing that capacities beyond this threshold yield diminishing returns, with cost increases outweighing the additional renewable integration benefits [55].

- *Residential Rooftop Photovoltaics.* Small-scale photovoltaic systems installed on residential building rooftops with individual capacity not exceeding 20 kW. The generation profile incorporates three orientations (south, south-west, and south-east) with 30° tilt angle, producing approximately 1350 full-load equivalent hours annually.
- *Non-Residential Rooftop Photovoltaics.* Medium-scale photovoltaic installations on commercial, industrial, and public building rooftops with capacity ranging from 20 kW to 1 MW. The generation profile assumes south orientation with 30° inclination, yielding approximately 1350 equivalent hours annually.
- *Ground-Mounted Photovoltaics with Single-Axis Tracking.* Large-scale solar installations exceeding 1 MW capacity equipped with single-axis tracking systems that follow the sun's daily path. These systems achieve approximately 1568 equivalent hours annually, 16% higher productivity than fixed-tilt installations.
- *Agrivoltaic Systems.* Elevated photovoltaic installations designed to permit continued agricultural activities beneath the panels, integrating energy production with farming. The generation profile combines fixed south-facing configurations (1350 equivalent hours) and single-axis tracking systems (1568 equivalent hours) in equal proportions, resulting in an average annual production. This averaging approach was adopted because observed agrivoltaic performance data remain very limited in the Piemonte region; the two configurations selected represent the most widely deployed agrivoltaic typologies and their combination provides a reasonable central estimate.
- *Onshore Wind Power.* Wind turbines installed at suitable locations identified through wind resource assessment and territorial analysis. The potential estimation combined current connection requests to the transmission system operator (Terna – Econnextion [64]) with technical potential data from literature [65]. Generation profiles derived from historical regional production yield approximately 1495 equivalent hours annually.
- *Electric Storage (Lithium-ion Batteries).* Grid-scale battery systems providing electricity storage for balancing supply and demand fluctuations. Preliminary simulations determined that capacity exceeding 500 GWh provides diminishing returns, the cost increase outweighs renewable integration benefits. Consequently, maximum potential was set at 100 GWh.
- *E-methane for Gas Grid Injection.* Synthetic methane (CH_4) produced through power-to-gas processes: electrolysis produces hydrogen from renewable electricity, which undergoes CO_2 hydrogenation and methanation to create synthetic natural gas injectable into existing gas networks. This enables sector coupling between electricity and gas systems.
- *Biomethane for Gas Grid Injection.* Upgraded biogas produced through two pathways: (1) anaerobic digestion of organic feedstocks (manure, agricultural residues, sewage, sequential crops, grazing residues, organic waste, sewage sludge), and (2) thermal gasification of lignocellulosic materials (forest residues, wood waste, municipal solid waste, pruning).
- *Electrolyzer Capacity for Hydrogen Generation.* Water electrolysis systems converting renewable electricity into hydrogen through electrochemical splitting. Hydrogen serves multiple functions: industrial feedstock, seasonal energy storage, potential fuel for high-temperature processes. Maximum capacity set at 20,000 MW (20 GW) based on projected renewable electricity surplus availability and system integration requirements.
- *Hydrogen Storage.* Infrastructure for storing hydrogen produced by electrolyzers, enabling temporal decoupling of production and consumption.
- *Heat Pumps for District Heating.* Large-scale electric heat pumps integrated into district heating networks, extracting low-temperature heat from ambient sources (air, water, ground, waste heat) and upgrading it to temperatures suitable for network distribution (70-90 °C). These systems substantially increase heating efficiency and enable renewable electricity integration into thermal systems. Potential estimated based on existing district heating infrastructure extent and thermal demand.
- *Thermal Storage for District Heating.* Hot water storage tanks integrated into district heating networks, typically large-volume insulated vessels storing water at network temperature. These systems provide operational flexibility: storing excess heat during off-peak hours or periods of renewable electricity surplus, releasing heat during peak demand or electricity scarcity.
- *Energy Efficiency of Individual Buildings.* Passive energy conservation measures including thermal insulation of roofs, foundations, and facades, plus high-performance window replacement in residential buildings. The building stock was spatially characterized and classified by typology (single-family houses, multi-family houses, apartment blocks) and construction period (nine periods from pre-1919 to post-2010), with specific energy demand profiles for each category [7]. A bottom-up methodology developed in previous work [66] derived energy saving cost curves correlating investment costs with achievable energy savings. The methodology follows these steps [7]:
  - I. Building stock analysis: Classification of regional residential building stock by construction period, housing type, and heating degree days.
  - II. Energy consumption evaluation: Assessment of specific heat consumption for each construction period, and housing type.
  - III. Renovation assessment: Simulation of representative building types (single-family houses, multi-family houses, apartment blocks) in pre-renovation and post-renovation conditions using Passive House Planning Package (PHPP) to quantify energy savings.
  - IV. Cost-effectiveness ranking: Calculation of investment cost per annual kWh saved for each renovation measure, arranged from most to least cost-effective.

The resulting cost curve shows that initial interventions (roof insulation for pre-1946 buildings, followed by façade and basement insulation) offer high energy savings relative to investment costs, while later interventions (e.g., window replacement in post-2005 buildings) exhibit lower cost-effectiveness [7]. This decision variable represents the percentage reduction in space heating demand achieved through building envelope improvements. The variable ranges from 0% (no renovations, baseline condition) to 70% (maximum feasible reduction through comprehensive retrofit). The 70% upper bound reflects realistic renovation potential identified through the cost curve analysis. Energy efficiency measures reduce space heating demand but do not affect domestic hot water (DHW) requirements, which are tracked separately in the model [7]. The costs associated with energy efficiency are calculated using the energy saving cost curve and added to the total annual system costs outside the EnergyPLAN simulation.

- *Heat Pumps for Individual Heating*. Residential-scale heat pumps providing space heating and domestic hot water to individual buildings, replacing conventional boilers (gas, oil, coal, biomass, electric resistance). This decision variable represents the percentage of renovated heating demand that transitions to heat pump systems. A critical constraint ensures heat pumps are only installed in buildings that have undergone energy efficiency improvements [7]. This assumption prevents inefficient heat pump installation in poorly insulated buildings where performance would be suboptimal.

The variable Heat Pumps for Individual Heating is bounded between 0% (current minimal penetration) and 100% (all renovated buildings install heat pumps). The optimization determines which renovated buildings should install heat pumps following a substitution hierarchy that maximizes emission reduction: heat pumps replace (1) coal boilers first, (2) oil boilers second, (3) electric resistance heaters third, (4) natural gas boilers fourth, and (5) biomass boilers last [7]. This prioritization reflects the emission intensity of displaced heating systems. The model separately accounts for domestic hot water demand. For buildings connected to district heating networks, DHW is supplied by the district heating system. For individual buildings, DHW demand continues to be met by existing systems unless Heat Pumps for Individual Heating are installed [7].

- *Industrial Heat Pumps ($T < 100^{\circ}C$)*. Industrial heat pumps supplying thermal energy to industrial processes requiring temperatures below 100 °C (washing, drying, pasteurization, space heating in industrial facilities). Systems achieve coefficient of performance (COP) of approximately 2.77. This decision variable, denoted as Industrial Heat Pumps (T < 100 °C) and expressed as percentage, represents the share of industrial low-temperature process heat demand supplied by heat pumps rather than fossil fuel combustion. The variable is bounded between 0% (baseline 2021: negligible heat pump penetration) and 100% (complete electrification of low-temperature industrial heat). The optimization determines cost-optimal deployment considering electricity prices, COP, and emission reduction benefits.
- *Electric Furnaces ($T > 100^{\circ}C$)*. Direct electric heating systems for industrial processes requiring temperatures exceeding 100 °C. This decision variable, denoted as Electric Furnaces (T > 100 °C) and expressed as percentage, represents the share of industrial high-temperature thermal demand met through direct electrification. The variable ranges from 0% (baseline: minimal industrial electrification) to 100% (complete direct electrification of high-temperature processes). The optimization balances electrification against hydrogen-based solutions for different process types and temperature requirements.
- *Hydrogen for Industry ($T > 100^{\circ}C$)*. Hydrogen combustion or utilization as chemical reducing agent for industrial processes requiring temperatures exceeding 100 °C and difficult to electrify directly (steel production, cement manufacturing, high-temperature chemical processes). This decision variable, denoted as Hydrogen for Industry (T > 100 °C) and expressed as percentage, represents the share of industrial high-temperature thermal demand supplied through hydrogen. The variable is bounded between 0% (baseline: no industrial hydrogen use) and 100% (complete hydrogen-based decarbonization of high-temperature processes). Efficiency and costs estimated from literature on hydrogen production (via electrolysis) and combustion/process integration. The optimization determines the complementary roles of Electric Furnaces (T > 100 °C) and Hydrogen for Industry (T > 100 °C) for different industrial applications.

4. Results

4.1. Expert-Based exploration results

Following the methodology described in Section 2.1, six critical electrification technologies were selected for the expert-based exploration, representing the primary decarbonization levers across the heating and industrial sectors of the Piemonte region: E-methane for gas grid injection (Synthetic Gas), heat pumps for district heating (Heat pumps DH capacity), heat pumps for individual heating (Heat pumps individual heating), industrial heat pumps for low-temperature processes (Industrial heat pumps), electric furnaces for high-temperature processes (Electric Furnaces), and hydrogen for high-temperature industrial processes (Industrial H_2 penetration). The selection reflects the sector-coupling logic of the Piemonte energy system, where electrification of thermal and industrial demand must be evaluated in relation to the progressive penetration of variable renewable energy sources in the electricity grid. The ranges explored for each variable are reported in Table 1. To evaluate each technology under favourable system conditions, a set of enabling infrastructure variables was fixed at maximum capacity throughout the simulations. Specifically: E-methane and Industrial H_2 are assessed with electrolyzer capacity and hydrogen storage at their upper bounds, to represent full Power-to-Gas (PtG) integration; Heat pumps DH is analyzed with thermal storage at maximum capacity, which allows temporal decoupling of heat production from renewable electricity availability; Heat pumps individual heating is combined with maximum building energy efficiency (70% heating demand reduction). Electric Furnaces and Industrial HP, as direct electrification technologies with no associated storage or intermediate conversion stage, are evaluated without additional infrastructure assumptions.

Results are presented using the two complementary sensitivity indicators defined in Equations (4) and (5): the $Relative\_CO_2\_change$ and the $Relative\_Costs\_change$, both normalized against the baseline 2021 scenario. For each technology, a pair of contour maps is produced, one for each RES penetration metric (RES_share^{el} and $RES\_share^{el,effective}$), enabling a comprehensive reading of both the overgeneration regime and the effective renewable integration regime. Figs. 1 and 2 present the $Relative\_CO_2\_change$ changes as a function of RES penetration and technology deployment, using the RES_share^{el} and $RES\_share^{el,effective}$ metrics, respectively. Thermally-driven electrification technologies such as heat pumps district heating capacity, heat pumps individual heating, industrial heat pumps, exhibit an immediate and monotonically increasing decarbonization potential, independent of the current renewable penetration in the grid. Power-to-Gas and direct high-temperature electrification technologies, Synthetic Gas to be injected in the gas grid, Electric Furnaces, and Industrial hydrogen, display a different dynamic. In Fig. 1, all three variables exhibit a clear threshold behaviour: below approximately 90% of RES_share^{el}, the introduction of these technologies results in an increase in net system CO_2 emissions

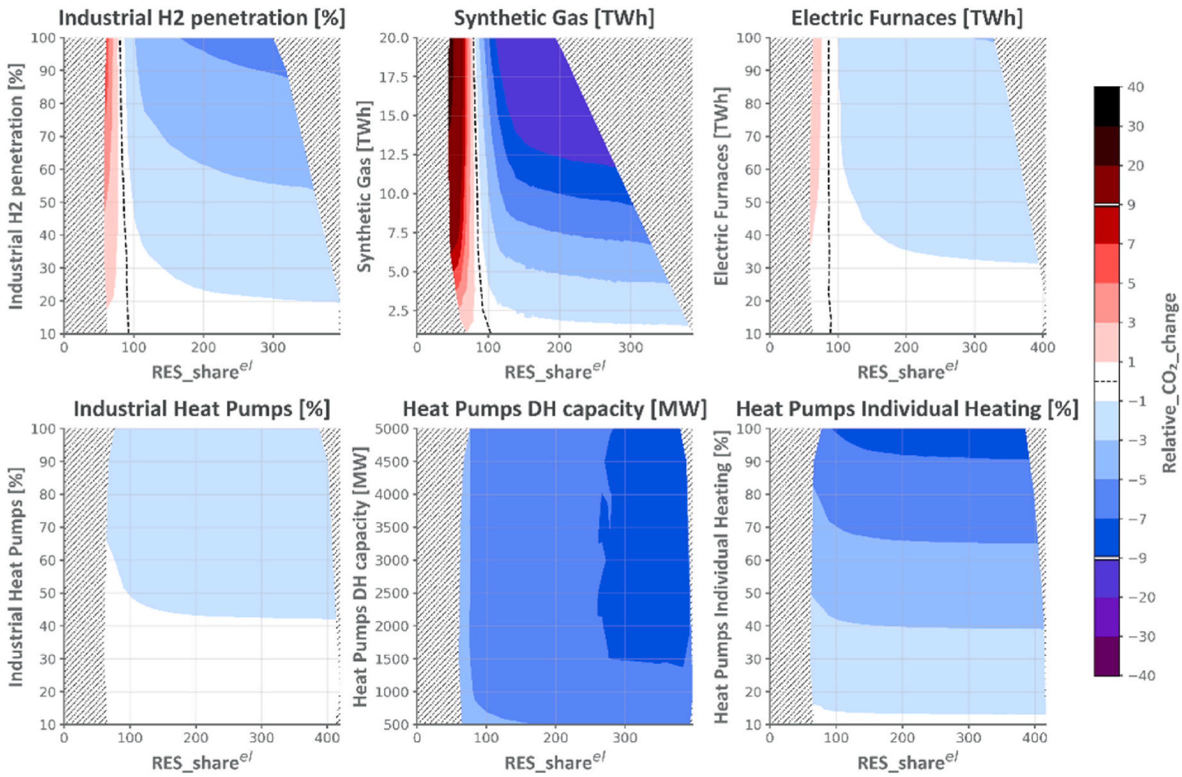

Fig. 1. Contour plots of *Relative_CO$_2$_change* [%] as a function of RES share of electric demand *RES_share^{el}* and technology penetration for six electrification technologies. Blue regions indicate net emission reductions; red regions indicate net increases relative to the no-technology scenario. Hatched areas denote technically non-feasible regions.

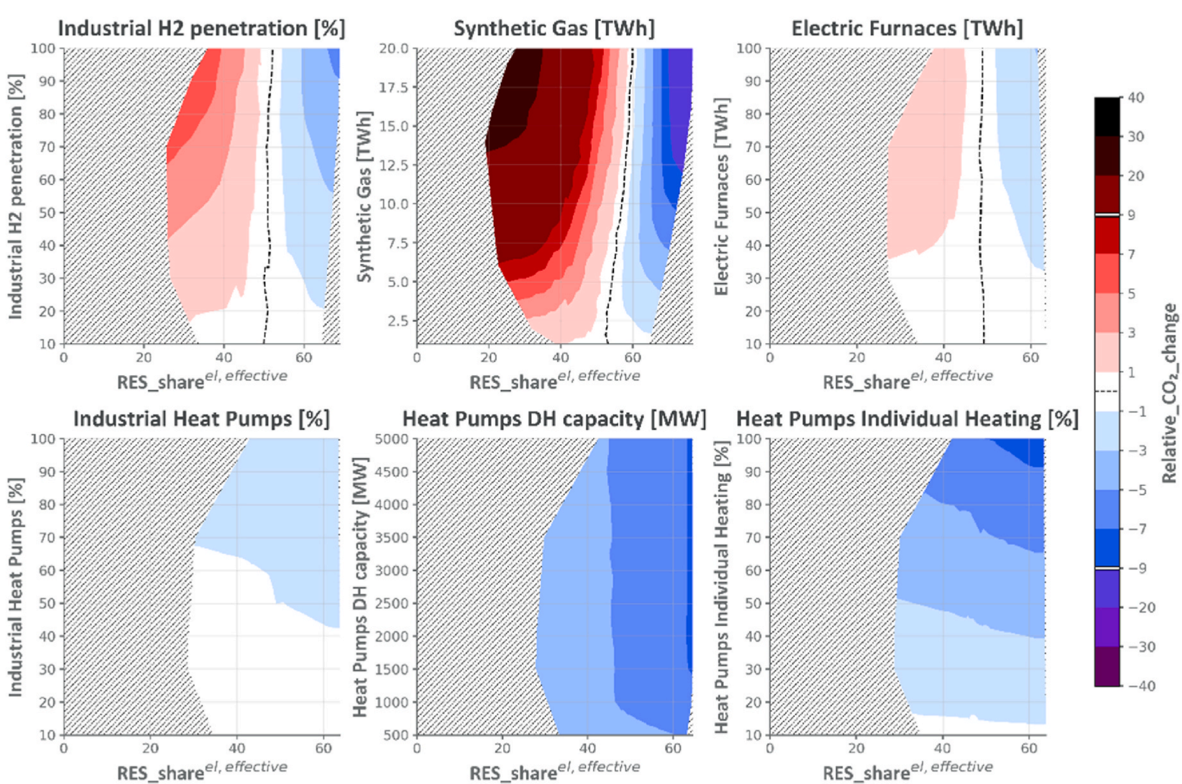

Fig. 2. Contour plots of *Relative_CO$_2$_change* [%] as a function of effective RES share of electric demand *RES_share^{el,effective}* and technology penetration for six electrification technologies. Blue regions indicate net emission reductions; red regions indicate net increases relative to the no-technology scenario. Hatched areas denote technically non-feasible regions.

relative to the reference case without the technology. This threshold at ~90% *RES_share*^{el} represents the tipping point beyond which the grid has been sufficiently decarbonized to unlock the environmental benefit of these technologies. This result carries substantial policy significance: premature large-scale deployment of Power-to-X or direct electrification technologies in hard-to-abate industrial sectors, in the absence of adequate renewable generation capacity, risks locking in additional net emissions rather than reducing them. The model therefore implicitly prescribes a sequencing logic for the regional energy transition: VRES expansion must precede or accompany, rather than follow, industrial Power-to-X deployment.

The use of the *RES_share*^{el,effective} metric in Fig. 2 corroborates these findings while sharpening the identification of the critical threshold. By excluding overgeneration from the RES accounting, this metric focuses exclusively on the fraction of demand covered by renewable sources at the moment of generation, filtering out periods of uncurtailed surplus that would otherwise inflate the apparent renewable share. Under this lens, the threshold shifts leftward to approximately 50% of *RES_share*^{el,effective}. This threshold is specific to the technology cost and grid carbon intensity assumptions of the Piemonte 2050 scenario. It would shift downward with a cleaner residual electricity mix or higher electrolyzer efficiency, and upward under more carbon-intensive grid conditions. It should therefore be interpreted as a case-specific indicative value rather than a transferable design target. The curvature and orientation of the contour isolines offer additional qualitative information. For Heat pumps DH capacity, the lines present a slight curvature toward higher RES values at intermediate technology deployment levels, indicating an emerging complementarity: at high RES penetration, thermal storage enables the heat pump to absorb surplus renewable electricity (Power-to-Heat), amplifying both the environmental and system flexibility benefits. For Synthetic Gas and Industrial H_2, the isolines in the beneficial region (above the threshold) steepen as RES approaches full system coverage, signalling that the marginal emission reduction per unit of technology deployment accelerates as the electricity mix approaches carbon neutrality.

Figs. 3 and 4 present the relative cost change as a function of RES penetration and technology deployment, using the *RES_share*^{el} and *RES_share*^{el,effective} metrics, respectively. The economic analysis reveals a more complex and technology-specific landscape than the emission analysis, with several distinct cost-response patterns emerging from the data. The introduction of synthetic gas generates a substantial increase in total annual system costs across the entire RES deployment range, driven by the high capital intensity of the electrolyzer and methanation plant infrastructure, the energy losses inherent in the multi-step conversion chain (electricity → hydrogen → synthetic methane), and the assumed CO_2 sourcing cost. At low to moderate RES penetration levels, the cost penalty is particularly severe, as the surrogate must operate primarily on conventionally generated electricity at market cost. As *RES_share*^{el} increases toward and beyond 100%, the rate of cost increase diminishes progressively: in the overgeneration regime, the system gains access to low- or zero-marginal-cost excess electricity that would otherwise be curtailed, improving the economics of the Power-to-Gas process and partially offsetting the capital cost burden. However, even in the most favourable RES conditions explored, Synthetic Gas remains a net cost-increasing technology, suggesting that it fulfils a role of system flexibility and long-duration storage rather than pure cost minimization. Electric Furnaces exhibit a qualitatively similar cost trajectory to Synthetic Gas, albeit with a lower absolute cost increment. This is attributable to the simpler single-step conversion process (electricity → heat), which eliminates the capital costs and energy losses of intermediate conversion stages. At high RES penetration, the cost increment associated with electric furnaces shrinks further, as the technology increasingly draws on surplus renewable electricity.

Technologies expressed as percentages of market saturation, heat pumps individual heating, industrial heat pumps, and industrial hydrogen, display a characteristic cost response that reflects the

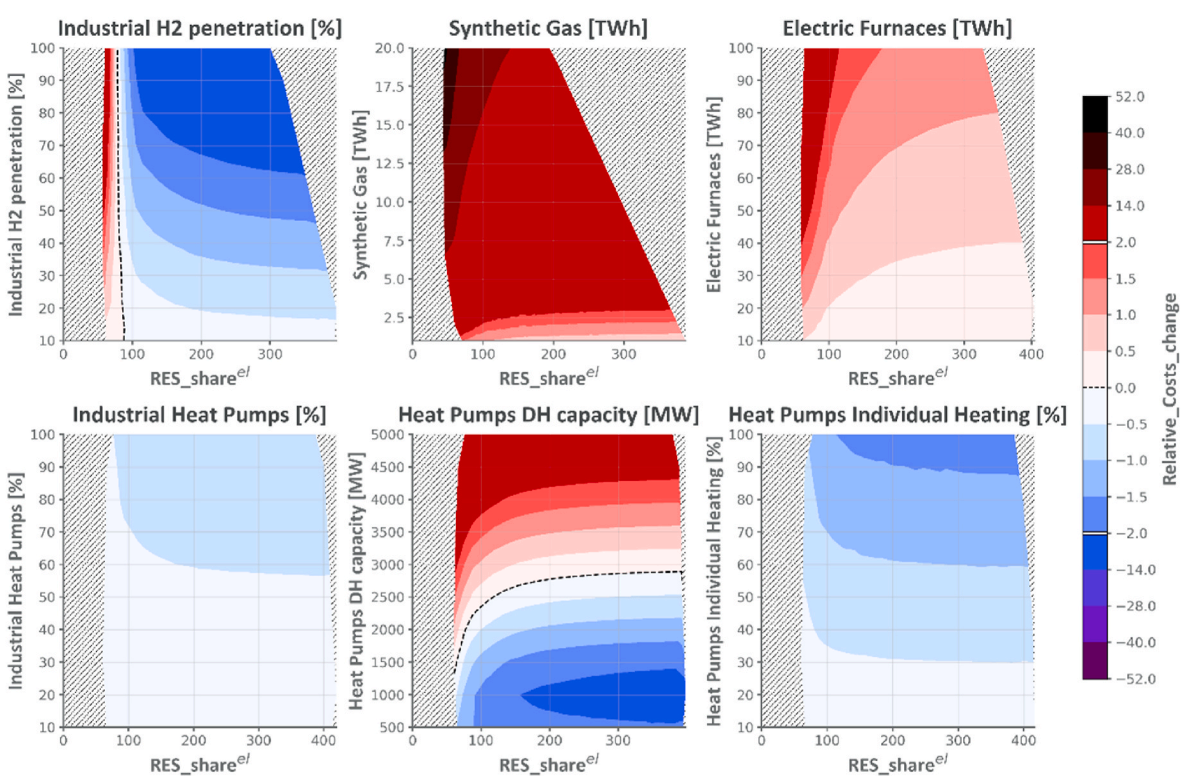

Fig. 3. Contour plots of *Relative_Costs_change* [%] as a function of RES share of electric demand *RES_share*^{el} and technology penetration for six electrification technologies. Blue regions indicate net emission reductions; red regions indicate net increases relative to the no-technology scenario. Hatched areas denote technically non-feasible regions.

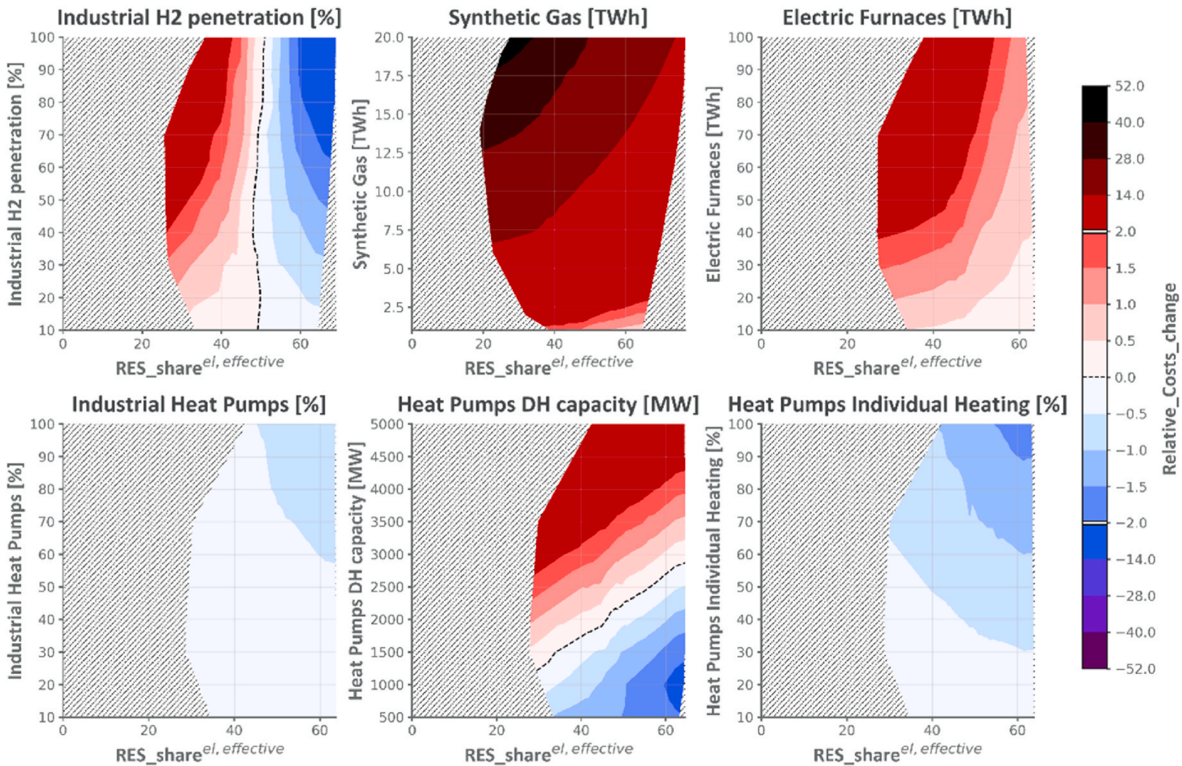

Fig. 4. Contour plots of *Relative_Costs_change* [%] as a function of effective RES share of electric demand *RES_share*^{el,effective} and technology penetration for six electrification technologies. Blue regions indicate net emission reductions; red regions indicate net increases relative to the no-technology scenario. Hatched areas denote technically non-feasible regions.

saturation nature of their formulation. For these variables, the maximum cost benefit is achieved simultaneously at maximum technology penetration and maximum RES share, corresponding to the upper-right corner of the contour map. The isolines for these variables are broadly diagonal in the decision space, confirming that cost and emission reductions are jointly achievable along a coordinated deployment trajectory that advances both RES expansion and end-use electrification in parallel. Heat pumps district heating, expressed in terms of absolute installed capacity (MW) rather than market saturation, reveals a distinctive non-monotonic cost response that distinguishes it from the other technologies. At low-to-moderate capacity levels, the cost response follows the same declining trajectory observed for the percentage-based variables, as each additional MW of heat pump capacity displaces fossil fuel combustion in the district heating network. However, beyond an optimal saturation capacity, visible in Fig. 3 as the point at which the cost contours begin to curve back, further capacity additions no longer reduce fossil fuel consumption, since the thermal demand of the network has already been met. Beyond this threshold, each additional MW of heat pump capacity contributes only to CAPEX without generating a corresponding emissions or fuel cost reduction, causing the total cost indicator to rise. This result highlights the critical importance of appropriately sizing district heating heat pump capacity against the actual thermal load of the network.

Fig. 4 confirms all the findings described above when using the *RES_share*^{el,effective} metric, while offering a cleaner visualization of the economic optimization trajectory. The elimination of overgeneration from the horizontal axis sharpens the distinction between technologies that are cost-effective even at low effective renewable shares (HP–Individual, Industrial HP) and those that require high effective renewable coverage to achieve acceptable cost performance (Synthetic Gas, Electric Furnaces).

4.2. Surrogate-enhanced Exploration analysis results

The development of the surrogate model followed a structured three-step selection process, documented in full in the Supplementary Material. First, a training dataset of 40,000 scenarios was generated by sampling the 17-dimensional decision space via Sobol sequences and simulating each configuration with EnergyPLAN v16.1. Second, a comprehensive benchmarking of 16 machine learning architectures was conducted, spanning linear models (Linear Regression, Ridge, Lasso), tree-based ensembles (Random Forest, Extra Trees, XGBoost, Gradient Boosting), and four neural network configurations (Small, Medium, Large, and Optimized), by training each candidate on progressively larger dataset fractions (40%, 60%, 80%) and evaluating performance through learning curve analysis, R^2 and RMSE on the held-out test set (20% of the data, never seen during training). Third, the best-performing architecture was selected based not only on pointwise accuracy but on Pareto front reproduction fidelity, before being deployed for the full surrogate-enhanced exploration.

The benchmarking revealed a clear three-tier hierarchy. Linear and instance-based models performed poorly ($R^2 < 0.85$), confirming the fundamental non-linearity of the energy system input–output mapping. Ensemble methods achieved intermediate accuracy (XGBoost: $R^2 = 0.985$), but their RMSE values remained an order of magnitude above the neural network tier. All four neural network variants achieved $R^2 > 0.999$, substantially outperforming all other model families. Among neural networks, the Optimized Neural Network achieved the highest pointwise R^2 (0.99944) but exhibited a clear saturation plateau in its learning curve from approximately 25,000 training samples onward, indicating that the architecture had exhausted the information extractable from the dataset. The Large Neural Network, by contrast, maintained a continuous improvement trajectory across all three training fractions ($R^2 = 0.99803 \rightarrow 0.99901 \rightarrow 0.99925$), demonstrating superior capacity to extract progressively deeper cross-sectoral structure from the data without reaching an information ceiling.

The decisive selection criterion was Pareto front reproduction fidelity, assessed by coupling each surrogate with the EPLANopt evolutionary algorithm and comparing the resulting Pareto approximation to the reference front obtained via full EnergyPLAN optimization. The Large Neural Network achieved GD = 0.012 and IGD = 0.012, outperforming the Optimized Neural Network (GD = 0.032, IGD = 0.034) by a factor of approximately 2.6–3.0 and XGBoost (GD = 0.065, IGD = 0.071) by a factor of approximately 5–6. This reversal, the model with the highest absolute R^2 producing the worst Pareto front among neural networks, directly demonstrates that learning saturation near the sparse Pareto region of the decision space penalizes architectures optimized for average accuracy, and that continuous learning dynamics translate into superior multi-objective fidelity where it matters most for energy planning. In physical units, the selected Large Neural Network achieves RMSE = 16.93 M€ on total annual costs on the held-out test set, confirming that individual scenario predictions deviate by less than 0.3% of the typical cost range across the full decision space. To verify that the Pareto front reproduction quality is not an artefact of a single optimization run, the EPLANopt algorithm was coupled to the Large Neural Network and executed across multiple independent runs with different random seeds, yielding stable mean convergence metrics (GD = 0.012, IGD = 0.012) with low variance across runs. This statistical robustness confirms that the surrogate produces consistent and reproducible optimization outcomes across different initialization conditions.

Fig. 5 presents the validation of the selected surrogate through two complementary representations. The upper panels shows the two Pareto fronts obtained with the EPLANopt and the deep-learning surrogate models. The subplots below show decision variable trajectories along the Pareto front in the two cases. Only 6 decision variables have been selected for graphical simplicity: residential rooftop PV, AgriPV, battery storage, synthetic gas, energy efficiency, and hydrogen electrolyzer capacity. The surrogate model replicates the trend of the real model, EnergyPLAN, with high fidelity, precisely capturing growth and turning points and ensuring that every point of the predicted front corresponds to a configuration physically coherent with EnergyPLAN's one. This high degree of fidelity certifies the surrogate model as a reliable tool for energy planning.

Fig. 6 presents the SHAP summary for annual CO_2 emissions (upper panel) total annual system costs (lower panel), ranking all 17 decision variables by their global importance and revealing both the magnitude and direction of each variable's marginal contribution across the sampled scenario space. AgriPV capacity emerges as the dominant driver of emission reduction, with a wide distribution of strongly negative SHAP values at high feature levels (red dots), suggesting that large-scale agrivoltaic deployment is a relevant lever in the surrogate's learned input-output mapping for the Piemonte system. The breadth of the distribution, extending to SHAP values below 4, indicates that AgriPV's contribution is not constant but highly state-dependent, amplifying as it interacts with storage and Power-to-X infrastructure. Energy efficiency improvement ranks second, with a consistent negative SHAP contribution at high penetration levels, reflecting the dual benefit of reduced thermal demand and the enabling condition it creates for heat pump deployment. Battery storage capacity and biomethane for industrial heating occupy intermediate positions, exhibiting larger SHAP magnitudes at high deployment levels, consistent with their role as enabling infrastructure for renewable integration. Critically, synthetic gas for grid injection, industrial electric furnaces, and industrial hydrogen heating display a distinctive bidirectional pattern: high feature values (red dots) are associated with positive SHAP values, that is, increased predicted emissions, while low feature values produce near-zero impact. This pattern is consistent with the threshold dynamics identified in Section 4.1: when deployed in a carbon-intensive electricity context, the upstream electricity consumption of Power-to-X technologies is likely to outweigh the emission savings at the end-use sector. The

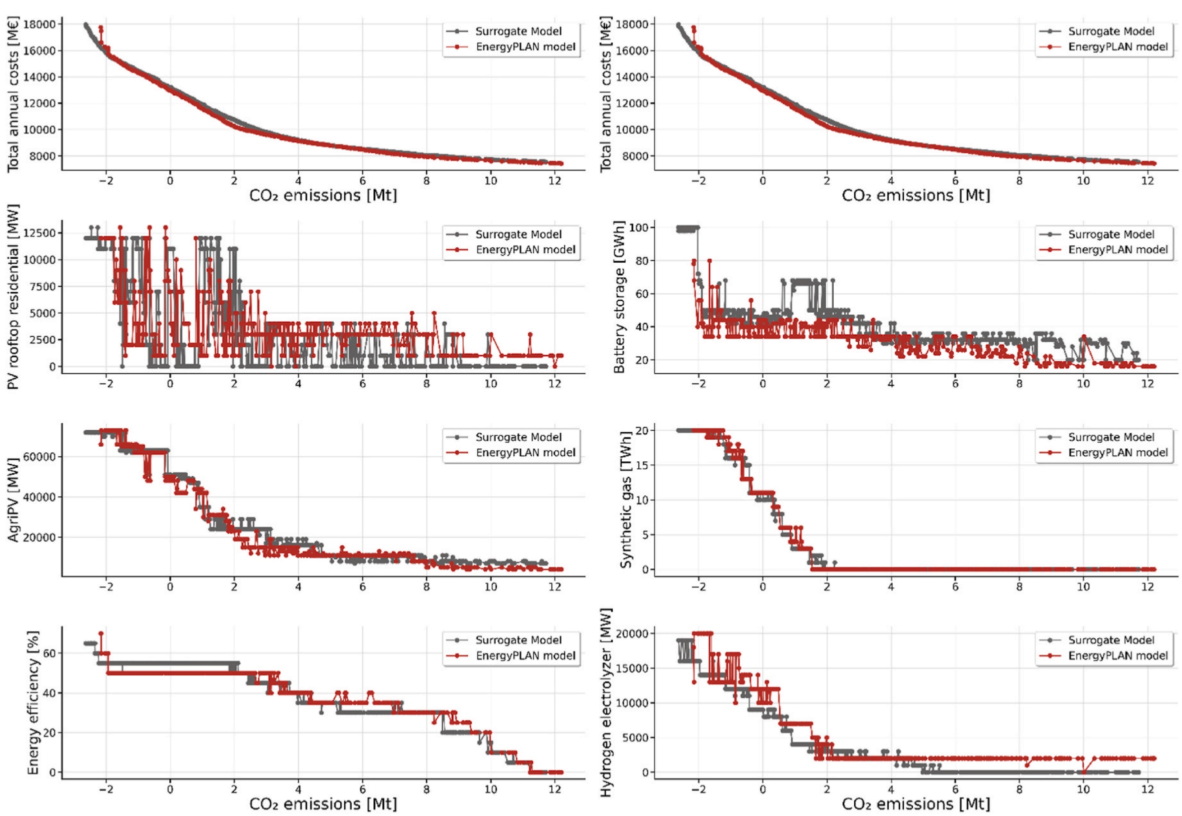

Fig. 5. Comparison of Pareto fronts and decision variable trajectories among the Pareto front solutions for the case obtained with EPLANopt and the case obtained through the deep learning surrogate model.

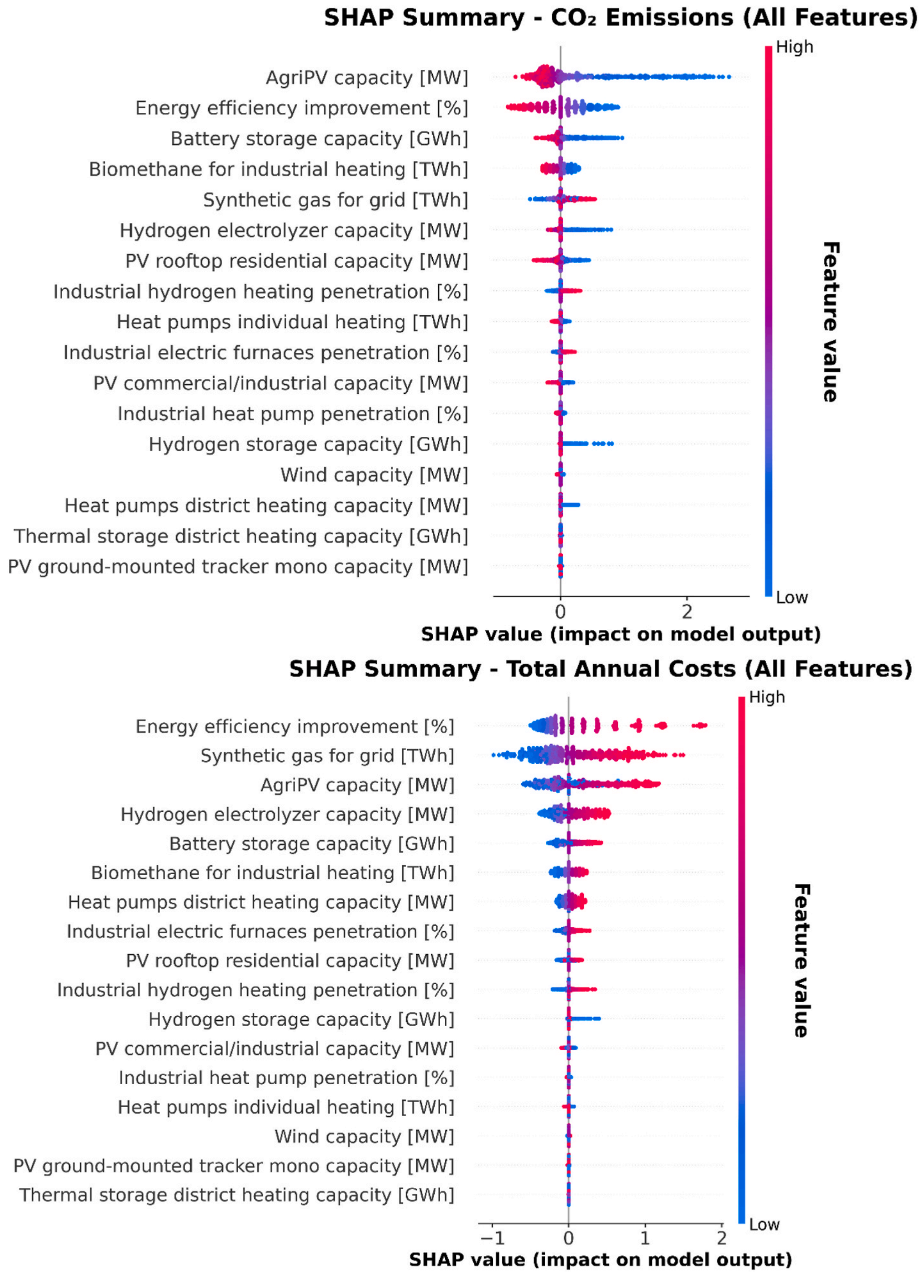

Fig. 6. SHAP summary plot for annual CO_2 emissions and total annual costs, ranking all 17 decision variables by global importance and directional cost impact.

surrogate appears to have internalized this conditional relationship, though SHAP values reflect the learned mapping rather than a direct verification of the underlying physical mechanism.

Within the cost ranking, energy efficiency improvement dominates the cost impact, with large positive SHAP values at high penetration levels, the capital intensity of comprehensive building retrofit programs is the primary cost driver across all scenarios. Synthetic gas for grid injection ranks second with large positive cost contributions, reflecting the multi-step conversion chain (electricity → hydrogen → synthetic methane) and the associated plant CAPEX. AgriPV capacity appears third in the cost ranking with predominantly positive SHAP values at high feature levels, indicating that large-scale agrivoltaics deployment increases total system costs if not coupled with flexibility options or electrification of other sectors. Heat pump technologies and industrial variables appear in the lower tier of the cost ranking with small SHAP magnitudes, reflecting their relatively low impact on total annual costs under 2050 technology assumptions.

Fig. 7 presents the Sobol interaction strength, the gap between total-order (S_T) and first-order (S_1) indices, for both CO_2 emissions and total annual costs across all 17 decision variables. For CO_2 emissions, AgriPV exhibits the highest interaction strength, indicating that a substantial fraction of its total influence on emissions is mediated through interactions with other technologies rather than through its direct, isolated contribution. The variables with which AgriPV interacts most strongly, synthetic gas and battery storage, are precisely those identified in the SHAP analysis as dependent on electricity surplus availability, consistent with the physical interpretation that AgriPV's contribution to emission reduction is amplified when paired with infrastructure able to absorb and valorise its generation surplus, though the Sobol indices characterize surrogate variance decomposition rather than directly measuring causal interdependence in the energy system. Synthetic gas ranks second in interaction strength, reflecting its fundamental dependence on renewable electricity availability and its cross-sectoral coupling with both PV generation and electrolyzer capacity. PV

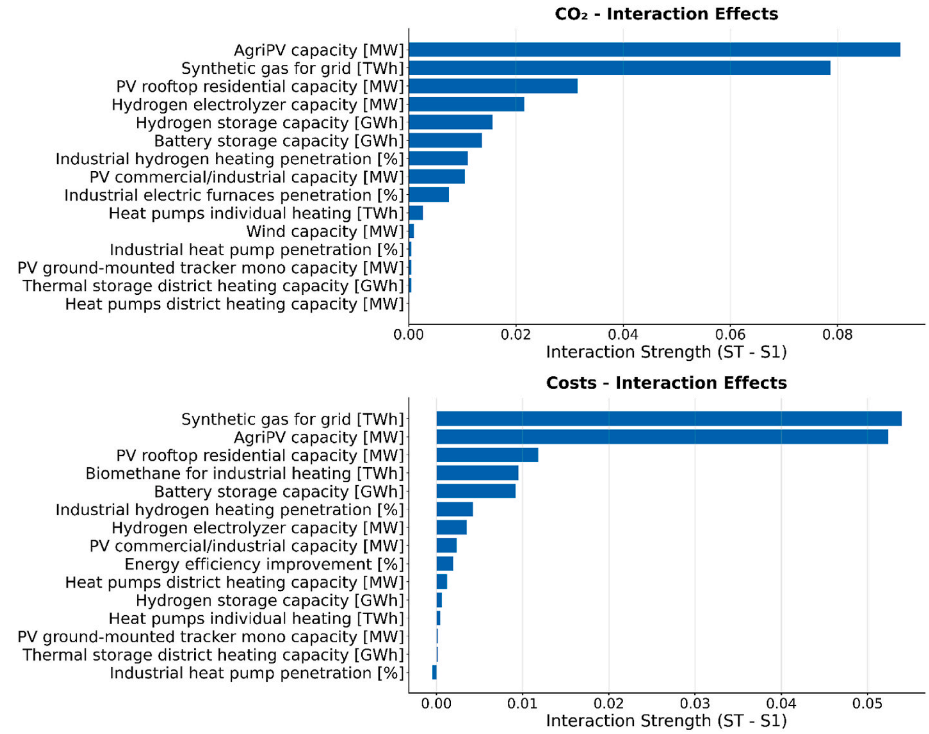

Fig. 7. Sobol interaction strength for both CO_2 emissions and total annual costs across all 17 decision variables. For CO_2 emissions.

rooftop residential, hydrogen electrolyzer, and hydrogen storage follow with moderate interaction strengths, consistent with their role as supporting infrastructure whose effectiveness scales with the available renewable surplus. At the lower end of the ranking, district heating heat pumps, thermal storage, and ground-mounted PV exhibit near-zero interaction strengths, confirming that these technologies deliver their emission reductions through direct substitution of fossil fuels and are largely independent of system-level context. For total annual costs, synthetic gas displays the highest interaction strength, followed by AgriPV and PV rooftop residential, with magnitudes generally lower than those observed for emissions, reflecting the more direct economic impact of these technologies through capital cost rather than through emergent system dynamics. The convergence between Sobol interaction leaders and SHAP importance leaders, with both methods identifying AgriPV and synthetic gas as the pivotal variables, strengthens confidence that the surrogate's learned structure is physically plausible and internally consistent.

Surrogate's computational efficiency enables exhaustive exploration of all 136 pairwise combinations of the 17 decision variables at high discretization resolution. For each decision variable, a matrix of heatmaps is generated showing the synergy emissions indicator $\Delta CO_2^{\text{synergy}}$ against every partner variable. To verify that the surrogate reproduces the local interaction structure and not only the global Pareto geometry, the full pairwise synergy matrix for Synthetic Gas was independently recomputed using a structured grid of full EnergyPLAN simulations. The resulting interaction patterns are qualitatively consistent with the surrogate-derived maps across all 16 variable combinations, with the AgriPV–Synthetic Gas pair confirmed as the dominant synergy in both cases; the full comparison is reported in the Supplementary Material.

Once the full pairwise synergy landscape is mapped for all variable combinations, the Interaction Impact Score is applied to rank all 136 pairs and identify the 16 with the strongest non-linear dynamics. Fig. 8 presents these top-ranked pairs. The results confirm that the dominant interaction structure is organized around the PV surplus–Power-to-Gas valorisation chain: AgriPV paired with synthetic gas, battery storage, and industrial hydrogen consistently achieves the largest synergy effects, while combinations involving PV rooftop residential with synthetic gas or PV commercial/industrial represent secondary interactions of the same physical chain at smaller scale. Pairs not involving a Power-to-X or storage variable exhibit near-zero synergy throughout, confirming that cross-sectoral surplus valorisation, rather than within-sector electrification, drives the emergent interaction dynamics of the Piemonte energy system.

Fig. 9 presents the third-order interaction isosurfaces for the highest-ranked cluster: AgriPV, battery storage, and synthetic gas. The isosurfaces exhibit pronounced and consistent curvature, with the most negative $\Delta CO_2^{\text{synergy}}$ values concentrated at the high-deployment corner of all three variables. This confirms that the three-way interaction is genuinely emergent and cannot be reduced to the sum of pairwise effects. A plausible physical interpretation links three sequential processes, solar generation, electrochemical storage, and Power-to-Gas conversion: battery storage may bridge the temporal mismatch between AgriPV generation peaks and electrolyzer demand, potentially maximising renewable utilization and synthetic gas output simultaneously. The isosurface structure is consistent with this mechanism. The full synergistic potential of the cluster is therefore only accessible when all three components are scaled in concert.

Fig. 10 presents the second-ranked cluster: AgriPV, synthetic gas, and hydrogen electrolyzer capacity. In contrast to Fig. 9, the isosurfaces exhibit pronounced flattening along the electrolyzer axis beyond approximately 10,000 MW. Below this threshold, each additional unit of electrolyzer capacity amplifies the AgriPV–synthetic gas synergy by converting a larger share of renewable surplus into synthetic methane. Beyond the threshold, the bottleneck shifts from electrolyzer capacity to the available electricity surplus itself, and further investment yields no additional synergistic benefit. This saturation threshold is not an intrinsic property of the electrolyzer technology but depends on the available renewable surplus in the cluster, which is itself a function of the AgriPV deployment level and battery storage capacity assumed in the analysis. A lower AgriPV deployment or reduced storage would shift the threshold proportionally downward. The 10 GW figure should therefore be read as an order-of-magnitude indicator for the Piemonte

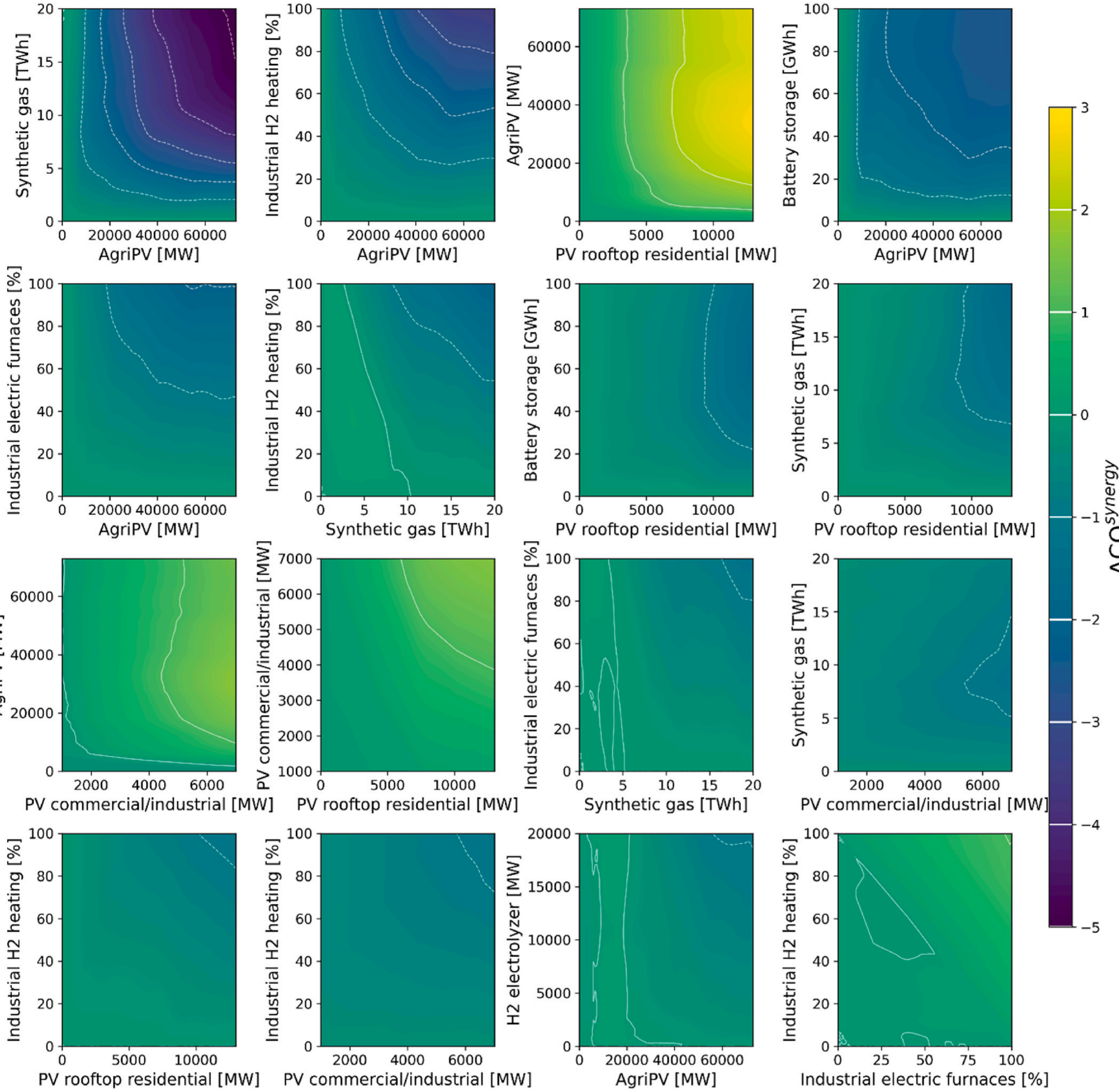

Fig. 8. Pairwise synergy emissions indicator ($\Delta CO_2^{synergy}$ [Mt]) for the 16 highest-ranked variable pairs selected from all 136 combinations by Interaction Impact Score.

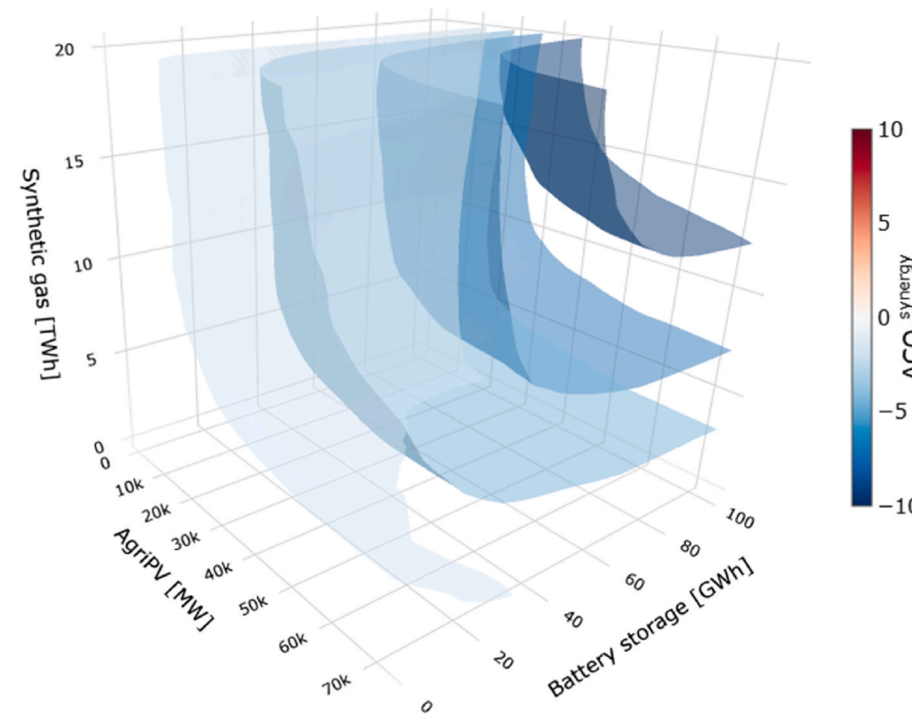

Fig. 9. Third-order interaction isosurfaces for the AgriPV–Battery storage–Synthetic gas cluster.

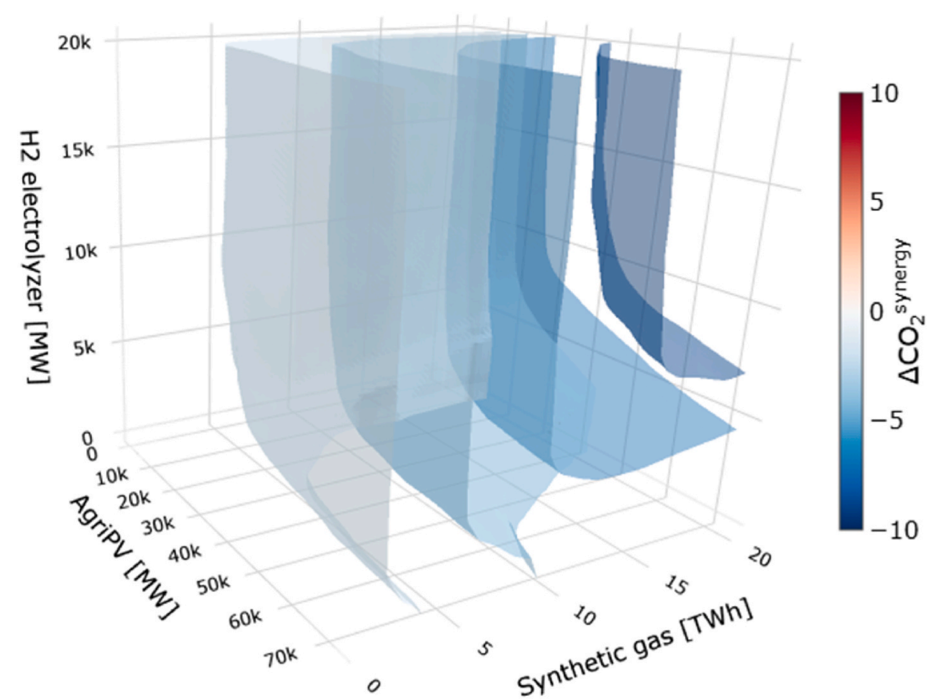

Fig. 10. Third-order interaction isosurfaces for the AgriPV–H2 electrolyzer–Synthetic gas cluster.

system under the assumptions of this cluster, not as a precise sizing recommendation.

5. Conclusions

The following conclusions are drawn from a single-node, hourly-resolution model of the Piemonte region calibrated to 2050 technology costs, demand projections, and regional policy targets. Results reflect the specific geographic and structural context of this system and should be interpreted as illustrative of interaction dynamics in highly industrialized European regional energy systems rather than as quantitatively transferable targets. This study developed and applied a two-stage methodology combining expert-based exploration and deep learning surrogate modeling to systematically identify, quantify, and interpret decision variable synergies in the multi-sectoral energy system of the Piemonte region of Italy for the target year 2050. The expert-based exploration revealed a fundamental behavioral dichotomy among electrification technologies. Heat pumps across all end-use sectors deliver net CO_2 reductions from the first unit of installed capacity and under any grid decarbonization scenario, making them robust early-stage instruments that should not be deferred. Conversely, Power-to-X technologies, synthetic gas, electric furnaces, and industrial hydrogen, require an effective renewable share exceeding approximately 50% before producing a net environmental benefit. Premature deployment in a carbon-intensive electricity context risks locking in additional net emissions, prescribing a clear sequencing logic in which renewable energy expansion must precede large-scale Power-to-X deployment.

The deep learning surrogate model trained on 40,000 EnergyPLAN simulations achieved $R^2 = 0.99925$ and accurately reproduced the reference EPLANopt Pareto front, demonstrating that surrogate modeling is a robust and computationally efficient alternative to full optimization for exploring the decision space. SHAP and Sobol-based XAI methods consistently identified agrivoltaic capacity and synthetic gas as the pivotal variables driving both emission reductions and cross-sectoral interactions, with results physically consistent with those of the expert-based stage, confirming the surrogate captures genuine energy system dynamics. Third-order interaction analysis demonstrated that battery storage acts as a critical mediator between PV generation and Power-to-Gas conversion, producing genuinely emergent synergies in the AgriPV–battery–synthetic gas cluster that cannot be decomposed into pairwise effects, and identifying an electrolyzer saturation threshold of approximately 10,000 MW beyond which co-deployment of storage becomes necessary to unlock further synergistic benefit. The methodology developed here is designed to be applicable to other regional and national energy planning contexts, though threshold values and interaction rankings will depend on the specific technology assumptions, demand structure, and resource availability of each system. Future work should extend the interaction analysis to higher-order clusters and multi-node spatial configurations.

CRediT authorship contribution statement

Matteo Giacomo Prina: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Carlo Pelizzoni:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Mackenzie Judson:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Madeleine McPherson:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Andrea Menapace:** Writing – review & editing, Supervision, Conceptualization. **Giampaolo Manzolini:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Wolfram Sparber:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and they take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of abbreviations

Nomenclature	
Total Annual Costs [*M€*]	Economic objective fuction
Annual CO_2 *Emissions* [*kt*]	Environmental objective fuction
$DV_k^{(L)}$	Lower bound of the k-th decision variable
$DV_k^{(U)}$	Upper bound of the k-th decision variable
DV_k	k-th decision variable
k	Decision variable index (k = 1, …, 17)
RES_share^{el}	RES share of electric demand: ratio of total annual renewable electricity generation to total annual electricity demand [%]
E^{RES}	Total annual electricity generated from all renewable energy sources [TWh]
D^{el}	Total annual electricity demand [TWh]
$RES\_share^{el,effective}$	Effective RES share of electric demand: fraction of demand actually served by renewable generation at each time step, excluding surplus [%]
E_i^{RES}	Electricity generated from renewable sources in time step i [TWh]
D_i^{el}	Electricity demand in time step i [TWh]
$Relative\_CO_2\_change$	Relative CO_2 emission change: marginal emission impact of deploying a technology normalized to the baseline scenario [%]
$Relative\_Costs\_change$	Relative total annual cost change: marginal cost impact of deploying a technology normalized to the baseline scenario [%]
CO_2^{with}	Annual CO_2 emissions of the system when the selected technology is deployed [Mt]
$CO_2^{without}$	Annual CO_2 emissions when the selected technology is not deployed, with renewable penetration held constant [Mt]
$CO_2^{baseline}$	Annual CO_2 emissions of the starting (2021 baseline) scenario [Mt]
$Costs^{with}$	Total annual system costs when the selected technology is deployed [M€]
$Costs^{without}$	Total annual system costs when the selected technology is not deployed, with renewable penetration held constant [M€]
$Costs^{baseline}$	Total annual system costs of the baseline scenario [M€]
R^2	Coefficient of determination: surrogate model accuracy metric measuring the proportion of variance in EnergyPLAN outputs explained by the surrogate [−]
m	Scenario index

(*continued on next page*)

(*continued*)

Nomenclature	
M	Total number of scenarios used for surrogate training or validation
y_m	Actual EnergyPLAN output value for scenario m (used as ground truth in surrogate validation)
$\hat{y}_m$	Surrogate model predicted output value for scenario m
$\bar{y}_m$	Mean of actual EnergyPLAN output values across all M validation scenarios
A	Set of solutions forming the surrogate-predicted Pareto front
$GD(A)$	Generational Distance: average Euclidean distance from solutions in the surrogate-predicted Pareto front A to the nearest point on the reference Pareto front Z
Z	Reference Pareto front solution set obtained via full EPLANopt multi-objective optimization
d_m^p	Euclidean distance from scenario solution m in A to the nearest point on the reference Pareto front Z
p	Norm order used in the GD and IGD formulations (p = 1)
$\hat{d}_m^p$	Euclidean distance from each reference point to the nearest surrogate prediction.
$\Delta CO_2^{synergy}$	CO_2 synergy indicator: deviation of the joint emission reduction from the sum of individual reductions, quantifying non-linear interaction [Mt]
CO_2^{group}	Annual CO_2 emissions when all J decision variables of the cluster are simultaneously deployed at their respective levels [Mt]
$CO_{2,j}$	Annual CO_2 emissions when only decision variable j is active at its current level, with all other variables at lower bounds [Mt]
j	Index of an individual decision variable within the cluster (j = 1, …, J)
J	Number of decision variables forming the interaction cluster
I_{score}	Interaction Impact Score: composite metric used to rank technology clusters by the strength and variability of their non-linear synergy effect [−]
σ	Standard deviation of the CO_2 synergy indicator across the discretized decision subspace [Mt]
Acronyms	
AgriPV	Agrivoltaic Photovoltaic
AI	Artificial Intelligence
CAPEX	Capital Expenditure
CO_2	Carbon Dioxide
COP	Coefficient of Performance
DHW	Domestic Hot Water
DV	Decision Variable
GD	Generational Distance
GIS	Geographic Information System
HP	Heat Pump
HP–DH	Heat Pump for District Heating
IGD	Inverted Generational Distance
MOEA	Multi-Objective Evolutionary Algorithm
PEAR	Piano Energetico Ambientale Regionale
PHPP	Passive House Planning Package
PtG	Power-to-Gas
PV	Photovoltaic
R^2	Coefficient of Determination
RES	Renewable Energy Sources
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
VRES	Variable Renewable Energy Sources
XAI	Explainable Artificial Intelligence

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.segy.2026.100246.

Data availability

Data are shared within the Supplementary Material.

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